Accelerated Gradient Method

STAT 525 - Topics in Computational Statistics

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Overview

Complexity Theory of Convex Optimization

Accelerated Gradient Method

Convergence Analysis

Conjugate Gradient Method

Gradient Descent Method

- Recall the gradient descent method can achieve convergence rate

$$f(x^{(k)}) - f(x^*) \leq \frac{2M\|x^0 - x^*\|^2}{k} \quad \Rightarrow \quad O\left(\frac{1}{\epsilon}\right)$$

  if f is convex and M-smooth and

$$f(x^{(k)}) - f(x^*) \leq \left(1 - \frac{m}{M}\right)^k \left(f(x^{(0)}) - f(x^*)\right) \quad \Rightarrow \quad O\left(\kappa \log \frac{1}{\epsilon}\right)$$

  if f is m-strongly convex and M-smooth and $\kappa = M/m$.
- Naturally, we want to know if it is possible to accelerate convergence and what is limit of first order method.

Complexity Theory of Convex Optimization

- To answer such question and study the limit of first order method, we can adopt a framework for complexity theory of first-order convex optimization, developed by Nemirovski, Yudin, and Nesterov et al.
- The first order linear method is defined as an algorithm that outputs a sequence $\{x^{(k)}\}$ such that

$$x^{(k+1)} \in x^{(0)} + \mathrm{Lin}\{\nabla f(x^{(0)}), \ldots, \nabla f(x^{(k)})\},$$

  where Lin means linear span.
- Clearly, gradient descent method is a first order linear method.

Lower Bound of Convex Case

- A fundamental result from Nesterov 's book: For any k, $1 \leq k \leq (n-1)/2$ and any $x^{(0)}$, there exists a convex and M-smooth function such that for any first order linear method, we have

$$f(x^{(k)}) - f(x^*) \geq \frac{3M\|x^0 - x^*\|^2}{32(k+1)^2}$$

- This theorem provides us a lower bound on the error that any linear first-order algorithm must incur. In other words, there is a fundamental difficulty that limits how fast we can optimize a convex function by a first order linear method.

Lower Bound of Strong Convex Case

- Another fundamental result from Nesterov 's book: For any $x^{(0)}$, m and κ, there exists a m-strongly convex and κm-smooth function such that for any first order linear method, we have

$$f(x^{(k)}) - f(x^*) \geq \frac{m}{2}\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{2k} \|x^0 - x^*\|^2$$

- This theorem provides us a lower bound for the strongly convex case.
- In statistics, we have a simialr framework "minimax optimality theory". If you are interested in this, read "Tsybakov, A.B., 2008. Introduction to nonparametric estimation".

Gradient Descent and Lower Bound

Gradient Descent	Lower Bound
Convex case	Convex case
$O\left(\frac{1}{\epsilon}\right)$	$O\left(\frac{1}{\sqrt{\epsilon}}\right)$
Strongly convex case	Strongly convex case
$O\left(\kappa \log \frac{1}{\epsilon}\right)$	$O\left(\sqrt{\kappa} \log \frac{1}{\epsilon}\right)$

- There is a clear gap between gradient descent and lower bound. Is the lower bound too optimistic? Or is the gradient descent not good enough?

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Accelerated Gradient Method

- It turns out that the gap is due to the issues in gradient descent.
  - GD focuses on reducing the objective value at each iteration, which might be too "microscopic".
  - GD might experience some abrupt changes.
- The gradient descent can be accelerated by exploiting information from history.

Accelerated Gradient Descent

- Convex case

$$O\left(\frac{1}{\sqrt{\epsilon}}\right)$$

- Strongly convex case

$$O\left(\sqrt{\kappa}\log\frac{1}{\epsilon}\right)$$

Lower Bound

- Convex case

$$O\left(\frac{1}{\sqrt{\epsilon}}\right)$$

- Strongly convex case

$$O\left(\sqrt{\kappa}\log\frac{1}{\epsilon}\right)$$

Heavy-ball Method

- The heavy-ball method is first introduced by B. Polyak

$$x^{(k+1)} = x^{(k)} - t^{(k)} \nabla f(x^{(k)}) + s^{(k)}(x^{(k)} - x^{(k-1)})$$

- The main difference from gradient descent method is the momentum term $(x^{(k)} - x^{(k-1)})$. The momentum term allows to imitate the motion of a heavy ball.
- Suppose f is m-strongly convex and M-smooth **quadratic** function. If we choose $t^{(k)} = 4/(\sqrt{m} + \sqrt{M})^2$ and $s^{(k)} = \max(|1 - \sqrt{t^{(k)}M}|, |1 - \sqrt{t^{(k)}m}|)^2$. Then

$$\|x^{(k)} - x^*\| \leq \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right)^k \|x^{(0)} - x^*\|.$$

Nestorov's Method

- To accelerate gradient descent for more general convex case, Nestorov propose a simple, but effective method

$$x^{(k+1)} = y^{(k)} - t^{(k)} \nabla f(y^{(k)})$$
$$y^{(k+1)} = x^{(k+1)} + \frac{k}{k+3}(x^{(k+1)} - x^{(k)})$$

- Nesterov, Y.E., 1983. A method for solving the convex programming problem with convergence rate $O(1/k^2)$. In Dokl. akad. nauk Sssr (Vol. 269, pp. 543-547).
- Nesterov, Y. and Nemirovsky, A., 1988. A general approach to polynomial-time algorithms design for convex programming. Report, Central Economical and Mathematical Institute, USSR Academy of Sciences, Moscow.

Nesterov's Method

$$x^{(k+1)} = y^{(k)} - t^{(k)} \nabla f(y^{(k)})$$
$$y^{(k+1)} = x^{(k+1)} + \frac{k}{k+3}(x^{(k+1)} - x^{(k)})$$

- Each iteration takes almost the same computational cost as gradient descent
- Not a descent method any more
- Suppose f is convex and M-smooth If $t^{(k)} = 1/M$, then

$$f(x^{(k)}) - f(x^*) \leq \frac{2M\|x^{(0)} - x^*\|^2}{(k+1)^2}$$

Nesterov's Method

- Another form of Nestorov's method for convex case

$$x^{(k+1)} = y^{(k)} - t^{(k)} \nabla f(y^{(k)})$$

$$y^{(k+1)} = x^{(k+1)} + \frac{s^{(k)} - 1}{s^{(k+1)}}(x^{(k+1)} - x^{(k)})$$

  where $s^{(0)} = 1$ and $s^{(k+1)} = (1 + \sqrt{1 + 4{s^{(k)}}^2})/2$.
- $(s^{(k)} - 1)/s^{(k+1)}$ is asymptotically equivalent to $k/(k+3)$

Nesterov's Method

- Nestorov's method for strongly convex case

$$x^{(k+1)} = y^{(k)} - t^{(k)} \nabla f(y^{(k)})$$
$$y^{(k+1)} = x^{(k+1)} + \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}(x^{(k+1)} - x^{(k)})$$

- Beck, A. and Teboulle, M., 2009. A fast iterative shrinkage thresholding algorithm for linear inverse problems. SIAM journal on imaging sciences, 2(1), pp.183-202.

Interpretation of Acceleration

- Despite of its beauty, the mechanism behind Nestorov's method has long been conceived as pure "algebraic trick" and remains mysterious. Many attempts have been made to interpret the algorithm from different angles.
- Here we provide an interpretation from the angle of differential equation.

Interpretation of Acceleration

- The continuous limit of gradient descent can be written as

$$\frac{dx}{dt} = -\nabla f(x).$$

  The differential equation has fixed points precisely when $\nabla f(x) = 0$.
- Instead of first-order differential equation, we can consider a second order one

$$\mu \frac{d^2x}{dt^2} = -\nabla f(x) - b\frac{dx}{dt},$$

  where $\nabla f(x) = 0$ is still a fixed point.

Interpretation of Acceleration

- If we adopt a simple finite-difference approximation

$$\mu \frac{x(t+\Delta t) - 2x(t) + x(t-\Delta t)}{(\Delta t)^2} \approx -\nabla f(x(t)) - b\frac{x(t+\Delta t) - x(t)}{\Delta t}$$

- Reorganizing these terms

$$x(t+\Delta t) = x(t) - \alpha \nabla f(x(t)) + \beta(x(t) - x(t-\Delta t))$$

  This is just heavy-ball method.

- Nestorov's method

$$x(t+\Delta t) = x(t) - \alpha \nabla f(x(t) + \beta(x(t) - x(t-\Delta t))) + \beta(x(t) - x(t-\Delta t))$$

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Convergence Analysis

- We focuses on the following sequence

$$x^{(k+1)} = y^{(k)} - t^{(k)} \nabla f(y^{(k)})$$
$$y^{(k+1)} = x^{(k+1)} + \frac{s^{(k)} - 1}{s^{(k+1)}}(x^{(k+1)} - x^{(k)})$$

  where $s^{(0)} = 1$ and $s^{(k+1)} = (1 + \sqrt{1 + 4s^{(k)^2}})/2$.

- Suppose f is convex and M-smooth. If $t^{(k)} = 1/M$, then

$$f(x^{(k)}) - f(x^*) \leq \frac{2M\|x^0 - x^*\|^2}{(k+1)^2}$$

Proof: The idea is to think the $x^{(k)}$ as a dynamical system (indexed by discrete time k) defined by the accelerated gradient update. Build a discrete-time version of "Lyapunov function" (energy function) to show convergence.

Step 1: We can show $s^{(k)} \geq (k+2)/2$.

- The proof can be completed by induction on k. Clearly, $s^{(0)} = 1 \geq (0+2)/2$. Suppose $s^{(k)} \geq (k+2)/2$ for k and we will prove $s^{(k+1)} \geq (k+3)/2$. By definition, we have

$$s^{(k+1)} = \frac{1+\sqrt{1+4{s^{(k)}}^2}}{2} \geq \frac{1+\sqrt{1+(k+2)^2}}{2} \geq \frac{k+3}{2}.$$

Proof: Step 2: Define a "Lyapunov function"

$$V^{(k)} = \|u^{(k)}\|^2 + \frac{2}{M}s^{(k-1)^2}(f(x^{(k)}) - f^*),$$

where $u^{(k)} = s^{(k-1)}x^{(k)} - (x^* + (s^{(k-1)} - 1)x^{(k-1)})$. We can show $V^{(k)}$ is a non-increasing sequence, i.e.

$$V^{(k+1)} \leq V^{(k)}$$

Proof: We first show

$$f(y^+) - f(x) \leq \frac{M}{2}\|x-y\|^2 - \frac{M}{2}\|x-y^+\|^2, \tag{1}$$

where $y^+ = y - t\nabla f(y)$.

- By the descent lemma and convexity, we have

$$\begin{aligned}
&f(y^+) + \frac{M}{2}\|x-y^+\|^2 \\
\leq &f(y) + \nabla f(y)^T(y^+ - y) + \frac{M}{2}\left(\|y^+ - y\|^2 + \|x-y^+\|^2\right) \\
= &f(y) + \nabla f(y)^T(x-y) + \frac{M}{2}\|x-y\|^2 \\
\leq &f(x) + \frac{M}{2}\|x-y\|^2.
\end{aligned}$$

Proof: Step 2: In (1), we take $x = (x^* - x^{(k)})/s^{(k)} + x^{(k)}$ and $y = y^{(k)}$ to get

$$\begin{aligned}
&f(x^{(k+1)}) - f((x^* - x^{(k)})/s^{(k)} + x^{(k)}) \\
\leq &\frac{M}{2}\|(x^* - x^{(k)})/s^{(k)} + x^{(k)} - y^{(k)}\|^2 - \frac{M}{2}\|(x^* - x^{(k)})/s^{(k)} + x^{(k)} - x^{(k+1)}\|^2 \\
= &\frac{M}{2{s^{(k)}}^2}(\|u^{(k)}\|^2 - \|u^{(k+1)}\|^2)
\end{aligned}$$

Since $f(\cdot)$ is convex, we have

$$f((x^* - x^{(k)})/s^{(k)} + x^{(k)}) \leq f(x^*)/s^{(k)} + (1 - 1/s^{(k)})f(x^{(k)}).$$

Combining above two inequalities yields

$$f(x^{(k+1)}) - f(x^*)/s^{(k)} - (1 - 1/s^{(k)})f(x^{(k)}) \leq \frac{M}{2{s^{(k)}}^2}(\|u^{(k)}\|^2 - \|u^{(k+1)}\|^2)$$

Proof: Step 2: By the definition $s^{(k+1)} = (1 + \sqrt{1 + 4{s^{(k)}}^2})/2$, we then have

$${s^{(k)}}^2(f(x^{(k+1)}) - f(x^*)) - {s^{(k-1)}}^2(f(x^{(k)}) - f(x^*)) \leq \frac{M}{2}(\|u^{(k)}\|^2 - \|u^{(k+1)}\|^2)$$

We can show

$$V^{(k+1)} \leq V^{(k)}$$

after reorganizing terms.

Proof: Step 3: We can complete the proof by using $V^{(k+1)} \leq V^{(k)}$.

- Since $V^{(k+1)} \leq V^{(k)}$,

$$V^{(k)} \leq V^{(1)} = \|x^{(1)} - x^*\|_2 + \frac{2}{M}(f(x^{(1)}) - f^*)$$

By taking $x = x^*$ and $y = y^{(0)} = x^{(0)}$ in (1), we have

$$f(x^{(1)}) - f(x^*) \leq \frac{M}{2}\|x^* - x^{(0)}\|^2 - \frac{M}{2}\|x^* - x^{(1)}\|^2$$

This suggests that

$$V^{(k)} \leq \|x^* - x^{(0)}\|^2$$

Proof: Step 3:

- $V^{(k)} \leq \|x^* - x^{(0)}\|^2$ leads to

$$\frac{2}{M} s^{(k-1)^2} (f(x^{(k)}) - f^*) \leq \|x^* - x^{(0)}\|^2$$

We can complete the proof by noting $s^{(k-1)} \geq (k+1)/2$.

Convergence Analysis

- We move to strongly convex case

$$x^{(k+1)} = y^{(k)} - t^{(k)} \nabla f(y^{(k)})$$
$$y^{(k+1)} = x^{(k+1)} + \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}(x^{(k+1)} - x^{(k)})$$

- Suppose f is m-strongly convex and M-smooth. If $t^{(k)} = 1/M$, then

$$f(x^{(k)}) - f(x^*) \leq \left(1 - \frac{1}{\sqrt{\kappa}}\right)^k \left(f(x^{(0)}) - f(x^*) + \frac{m\|x^0 - x^*\|^2}{2}\right)$$

Proof: The proof itself is very similar to convex case.

Step 1: Define

$$V^{(k)} = \frac{m}{2}\|u^{(k)}\|^2 + (f(x^{(k)}) - f^*),$$

where $u^{(k)} = \sqrt{\kappa}x^{(k)} - (x^* + (\sqrt{\kappa} - 1)x^{(k-1)})$. We can show $V^{(k)}$ is a contraction sequence, i.e.

$$V^{(k+1)} \leq \left(1 - \frac{1}{\sqrt{\kappa}}\right) V^{(k)}$$

We leave the proof as exercise.

Proof: Step 2: We can know that

$$V^{(k)} \leq \left(1 - \frac{1}{\sqrt{\kappa}}\right)^k \left(f(x^{(0)}) - f(x^*) + \frac{m\|x^0 - x^*\|^2}{2}\right)$$

We can complete the proof by noting $f(x^{(k)}) - f(x^*) \leq V^{(k)}$.

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Conjugate Gradient Method

- In the strongly convex quadratic function, we can also introduce another momentum method, called conjugate gradient method.
- Consider the following problem

$$f(x) = \frac{1}{2}x^T Q x - b^T x + c,$$

where Q is positive definite matrix.

Conjugate Gradient Method

- We can consider the formulation of heavy-ball method

$$x^{(k+1)} = x^{(k)} - t^{(k)} \nabla f(x^{(k)}) + \beta^{(k)} (x^{(k)} - x^{(k-1)})$$

and introduce some notations

$$x^{(k+1)} = x^{(k)} + t^{(k)} p^{(k)} \quad \text{and} \quad r^{(k)} = \nabla f(x^{(k)}) = Q x^{(k)} - b.$$

- With these notations, we can find

$$p^{(k)} = -r^{(k)} + \gamma^{(k-1)} p^{(k-1)}$$

and

$$r^{(k+1)} = Q x^{(k+1)} - b = Q x^{(k)} - b + t^{(k)} Q p^{(k)} = r^{(k)} + t^{(k)} Q p^{(k)}.$$

Conjugate Gradient Method

- Now we can have a new expression of momentum

$$
\begin{aligned}
x^{(k+1)} &= x^{(k)} + t^{(k)} p^{(k)} \\
r^{(k+1)} &= r^{(k)} + t^{(k)} Q p^{(k)} \\
p^{(k+1)} &= -r^{(k+1)} + \gamma^{(k)} p^{(k)}
\end{aligned}
$$

- Different from heavy-ball method, we choose $t^{(k)}$ and $\gamma^{(k)}$ based some new criteria.
  1. We choose $t^{(k)}$ by exact line search

$$
t^{(k)} = \frac{(p^{(k)})^T r^{(k)}}{(p^{(k)})^T Q p^{(k)}}
$$

  2. We choose $\gamma^{(k)}$ to ensure conjugacy, i.e. $(p^{(k)})^T Q p^{(k+1)} = 0$

$$
\gamma^{(k)} = \frac{(r^{(k+1)})^T Q p^{(k)}}{(p^{(k)})^T Q p^{(k)}} = \frac{(r^{(k+1)})^T r^{(k+1)}}{(r^{(k)})^T r^{(k)}}.
$$

Conjugate Gradient Method

- The search directions in conjugate gradient method are conjugate to each other
$$(p^{(k_1)})^T Q p^{(k_2)} = 0, \qquad k_1 \neq k_2.$$
- $x^{(k+1)}$ is the minimizer of f in $x^{(0)} + \text{span}\{p^{(0)}, \ldots, p^{(k)}\}$. So we can find minimizer in at most iterations.
- There are many extensions to nonlinear functions. They usually perform well, but theoretical understanding is still limited.